MILITARY SPECIFICATION

FUZES, ROCKET, M423 AND M427
LESS BOOSTER AND BOOSTER LEAD

This Amendment forms a part of Military Specification MIL-F-50548A (AR), dated 9 November 1978, and is approved for use by the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 3

3.3.1: Delete in its entirety, and substitute the following:

"3.3.1 Arming Time - The rotor assembly in the loaded S&A shall rotate from the fully safe position to the fully armed position in not less than 0.67 sec. and not more than 0.88 sec. when subjected to arming acceleration of 27.00 plus or minus 0.25 g's. The S&A shall meet this requirement at any temperature between -65°F and +165°F."

3.4.1: Delete in its entirety and substitute the following:

"3.4.1 Arming Time - The fuze assembly shall arm in not less than 0.63 sec. or more than 0.92 sec. when subjected to arming acceleration of 27.00 plus or minus 0.25 g's and the detent shall lock the rotor in the fully armed position, as determined by radiographic inspection. The fuze assembly shall meet this requirement at any temperature between -65°F and +165°F."

3.5.1: Delete in its entirety and substitute the following:

"3.5.1 Arming Time - The rotor assembly in the loaded S&A shall rotate from the fully safe position to the fully armed position in not less than 1.13 sec. and not more than 1.37 sec. when subjected to arming acceleration of 40.00 plus or minus 0.25 g's. The S&A shall meet this requirement at any temperature between -65°F and +165°F."

FSC 1340



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PAGE 4

3.6.1: Delete in its entirety and substitute the following:

"3.6.1 Arming Time - The fuze assembly shall arm in not less than 1.07 sec. and not more than 1.46 sec. when subjected to arming acceleration of 40.00 plus or minus 0.25 g's and the detent shall lock the rotor in the fully armed position as determined by radiographic inspection. the fuze assembly shall meet this requirement at any temperature between -65°F and +165°F.

* 3.9: Delete paragraph number and substitute "3.9.1". Add new paragraph 3.9:

"3.9 Drop tests"

Add new paragraph 3.9.2:

"3.9.2 Five-foot drop. The fuze shall be safe to handle, fire, or dispose of following the five-foot drop test as specified in MIL-STD-331. It need not be operable."

* 3.10: Delete after "tests": "Five Foot Drop Test as specified in MIL-STD-331;".

PAGE 5

* 3.10, line 5: Delete "the Five Foot Drop Test and".

* 4.1: Delete in its entirety and substitute:

"4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification."



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PAGE 6

4 3.1: Delete "253" and subsitute "376".

PAGE 7

* 4.3.1: Add "or 9260776" next to "8887706". Delete "8883693" and substitute "9260804". Add "or 9260774 next to "8883692".

Add "Stud, Spring 8883721 10".

PAGE 11

TABLE 1: Delete "253" and substitute "376"; under Fuze, Rocket, Less Booster and Booster Lead, add;

| EXAMINATION OR TEST | NUMBER OF SAMPLE UNITS | REQUIREMENTS PARAGRAPH | PARAGRAPH REFERENCE |
|---|---|---|---|
| Low temperature MIL-STD-810 Procedure 1, Method 502 | 73 | 3.4.1 | 4.5.35 |
| High temperature MIL-STD-810 Procedure 1 Method 501 | 73 | 3.4.1 | 4.5.36 |

PAGE 12

TABLE 1: Delete "253" and substitute "376"; under Fuze, Rocket, Less Booster and Booster Lead, add;

| EXAMINATION OR TEST | NUMBER OF SAMPLE UNITS | REQUIREMENTS PARAGRAPH | PARAGRAPH REFERENCE |
|---|---|---|---|
| Low temperature MIL-STD-810 Procedure 1, Method 502 | 73 | 3.6.1 | 4.5.35 |
| High temperature MIL-STD-810 Procedure 1 Method 501 | 73 | 3.6.1 | 4.5.36 |



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PAGE 15

TABLE 1: Delete "(Dwg. 8883706)" and substitute "(Dwg 8883706 or 9260776)".

PAGE 16

TABLE 1: Delete "8883693" and substitute "9260804".

Delete "(Dwg. 8883692)" and substitute "(Dwg 8883692 or 9260774)".

* Add the following:

"Stud, Spring
(Dwg 8883721)
Examination for defects 10 3.2 4.4.2.37"

PAGE 37

* 4.4.2.16: Delete Note a in its entirety and substitute:

"a. Samples taken from 5 foot drop, aircraft vibration, non-environmental (including low temperature and high temperature): Note 5' drop not done on every lot."

Add:

| CATEGORY | EXAMINATION OR TEST | NUMBER OF SAMPLE UNITS | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE |
|---|---|---|---|---|
| CRITICAL | | | | |
| 10 | Low Temperature MIL-STD-810, Procedure 1, Method 502 | 73 | 3.4.1 | 4.5.35 |
| 11 | High Temperature MIL-STD-810 Procedure 1, Method 501 | 73 | 3.4.1 | 4.5.36 |



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* 4.4.2.17: Delete Note a in its entirety and substitute: "a. Samples taken from 5 foot drop, Aircraft Vibration, Non-Environmental (including Low Temperature and High Temperature): Note 5' drop not done on every lot."

Add:

| CATEGORY | EXAMINATION OR TEST | NUMBER OF SAMPLE UNITS | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE |
|---|---|---|---|---|
| CRITICAL | | | | |
| 10 | Low Temperature MIL-STD-810, Procedure 1, Method 502 | 73 | 3.6.1 | 4.5.35 |
| 11 | High Temperature MIL-STD-810 Procedure 1, Method 501 | 73 | 3.6.1 | 4.5.36 |

PAGE 41

4.4.2.20, Major M114: Delete "visual" and substitute "gage".

PAGE 49

* 4.4.2.27: Under "CATEGORY" delete "1" and "2" and substitute "2" and "1", respectively.

PAGE 50

* 4.4.2.28: Under "CATEGORY" delete "1" and "2" and substitute "2" and "1", respectively.

PAGE 51

* 4.4.2.29: Under "CATEGORY" delete "1" and "2" and substitute "2" and "1", respectively.



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PAGE 52

* 4.4.2.30: Delete in its entirety and substitute included Page 7 (Page 52 of 81).

PAGE 53

* 4.4.2.31: Delete in its entirety and substitute included Page 8 (Page 53 of 81).

PAGE 54

* 4.4.2.32: Delete in its entirety and substitute included Page 9 (Page 54 of 81).

PAGE 56

4.4.2.34: Add "or 9260774" next to "8883692".

PAGE 62

* 4.5.5: Add "One defective shall be cause for lot rejection".

* 4.5.6: Add "One defective shall be cause for lot rejection".

* 4.5.7.1: Delete "Apply an" and substitute "Apply the required".

Add: "and removed from the lot".

* 4.5.7.2: Delete "Apply an" and substitute "Apply the required".

Add: "and removed from the lot".

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* 4.5.7.3: Delete "Apply an" and substitute "Apply the required".

Add "and removed from the lot".

* 4.5.7.4: Delete "Apply the" and substitute "Apply the required".

Add: "and removed from the lot".



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* 4.5.7.5: Delete "Apply an" and substitute "Apply the required".

Add: "and removed from the lot".

4.5.7.6, line 5: Delete "to the outside face of the annular gear" and substitute "to the outside face of the annular gear studs".



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## CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50548A (AR)

| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER 8883706 or 9260776 |
|---|---|---|---|---|---|
| 4.4.2.30 | Lever Assembly | | | | NEXT HIGHER ASSEMBLY 8883692 or 9260774 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Critical | | | | | |
| 1 | Maximum gap between lever and shaft | | 100% | 3.2 | Gage |
| 2 | Load test of lever shaft | | 100% | 3.2 | 4.5.7.4 |
| 3 | Load test of lever pallet | | 100% | 3.2 | 4.5.7.5 |
| Major | | | | | |
| 101 | Torque test of lever shaft | | 100% | 3.2 | 4.5.7.8 |
| Minor | | | | | |
| 201 | Evidence of poor workmanship | | 0.65% | 3.16 | Visual |

NOTES:

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## CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50548A (AR)

| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER 8883744 |
|---|---|---|---|---|---|
| 4.4.2.31 | Rotor Assembly | | | | NEXT HIGHER ASSEMBLY 9215619 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Critical 1 | Load test of annular gear | | 100% | 3.2 | 4.5.7.6 |
| Major 101 | Load test of lock roller pin | | 0.40% | 3.2 | 4.5.7.7 |
| Minor 201 | Evidence of poor workmanship | | 0.65% | 3.16 | Visual |

NOTES:

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## CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50548A (AR)

| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER 9260804 |
|---|---|---|---|---|---|
| 4.4.2.32 | Inner Plate Assembly | | | | NEXT HIGHER ASSEMBLY 8883692 or 9260774 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Critical 1 | None defined | | | | |
| Major 101 | Load test of bridge rivet | | 0.40% | 3.2 | 4.5.7.9 |
| Minor 201 | Evidence of poor workmanship | | 0.65% | 3.16 | Visual |

NOTES:

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## CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50548A (AR)

| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER 8883721 |
|---|---|---|---|---|---|
| 4.4.2.37 | Stud, Spring | | | | NEXT HIGHER ASSEMBLY 8883719 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Critical 1 | None defined | | | | |
| Major 101 | Diameter of stud, min | | 0.40% | 3.2 | Gage |
| Minor 201 | Evidence of poor workmanship | | 0.65% | 3.16 | Visual |

NOTES:

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* 4.5.7.8: Add "and removed from the lot".

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* TABLE II: Delete "Firing Test" in its entirety.

4.5.29: Delete in its entirety and substitute:

"4.5.29 Non-Environmental Test - A sample of two hundred and nineteen (219) fuzes shall be selected and subjected to and pass the non-arm test and acceptance criteria specified in 4.5.31. If the test is successful, seventy-three (73) of the 219 fuzes shall be randomly selected and subjected to and pass the arming test and acceptance criteria specified in 4.5.31. The remaining 146 fuzes shall be equally divided into two groups of 73 each. One group shall be subjected to the Low Temperature Test specified in 4.5.35 and the remaining group shall be subjected to the High Temperature Test specified in 4.5.36."

Add:

"4.5.35 Low Temperature Test - A sample of 73 fuzes shall be selected as specified in 4.5.29 and subjected to low temperature environment in accordance with Low Temperature Procedure I at -65°F, Method 502 of MIL-STD-810 with the exception that the time of exposure to temperature shall be a minimum of 4 hours. The fuzes shall then be subjected to and pass arming time test and acceptance criteria specified in 4.5.31 within 5 minutes of removal from Low Temperature.

NOTE:

Removing the fuze from the cold temperature environment prior to performing the centrifuge test may cause frost to form on the metal parts. This may have an effect on the test results due to attenuation of the acoustic signal of the arming mechanism, thus making it difficult to distinguish the arm signal from background noise. The contractor shall take appropriate action to alleviate this potential problem. One effective method has been to seal each fuze in a plastic bag prior to temperature conditioning. Bagged fuzes are then subjected to the centrifuge test.



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4.5.36 High Temperature Test - A sample of 73 fuzes shall be selected as specified in 4.5.29 and subjected to high temperature environment in accordance with High Temperature Procedure I, Method 501 of MIL-STD-810 with the exception that the time of exposure to temperature shall be a minimum of 4 hours. The fuzes shall then be subjected to and pass arming time test and acceptance criteria specified in 4.5.31 within 5 minutes of removal from High Temperature."

PAGE 71

4.5.32: Delete subparagraph a in its entirety and substitute "a. Times under .63 seconds for the M423 and 1.07 seconds for the M427 are classed as arming time under minimum."; Delete subparagraph b in its entirety and substitute "b. Times .93 seconds through 1.30 seconds for the M423 and 1.47 seconds through 1.52 seconds for the M427 are classed as arming time over maximum."

PAGE 72

4.5.32: Delete subparagraph c in its entirety and substitute "c. Times over 1.30 for the M423 and 1.52 for the M427 or fuzes which do not arm or fuzes which arm but do not detent are classified as failure to arm."; Delete subparagraph e in its entirety and substitute "e. All fuzes which are not classed as a failure shall be radiographically inspected prior to firing (vibrated fuzes) or their return to the lot (non-environmental fuzes including hot and cold fuzes)."

PAGE 74

6.3: Delete "DRDAR-QAR-I" and substitute "DRSMC-QAF-I(D)".

6.5: Delete "DRDAR-QAM-M" and substitute "DRSMC-QAF-T(D)".

Delete "DRDAR-LCF" and substitute "DRSMC-LCN-M(D)".

* 6.8: Delete in its entirety and substitute:

"6.8 When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified



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below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9(n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

| Paragraph No. | Data Requirement Title | Applicable DID No. |
|---|---|---|
| 4.5.34 | Quality Inspection Test, Demonstration and Evaluation Report | DI-R-1724 |
| 4.5.34 | Quality Inspection Defect Report | DI-R-1721 |

"(Data item descriptions related to this specification, and identified in Section 6 will be approved and listed as such in DoD 5000.19L., Vol II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)"

* "6.9 Source control and specification control drawings. Certificates of conformance from the supplier certifying that the item complies with the drawing requirements shall be provided to the Government representative."

*The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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Preparing activity:
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(Project 1340-A656)



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